

8:30 to 9:30 Classes

Leverage Asymmetric Multiprocessing and AI at the Edge for Scalable Embedded Systems **Supplier: Critical Link**

This seminar offers a technical discussion and live demonstration on two key approaches to modern embedded system design: AI at the edge and Asymmetric Multiprocessing (AMP) on multi-core platforms. Attendees will explore the benefits, challenges, and practical implementation strategies for both techniques, showcasing how Critical Link System-on-Modules (SOMs) — spanning Altera, Qualcomm, and other platforms — can accelerate scalable, intelligent embedded designs.

High-Power Ferrite Suppression and Efficiency Optimization in Wireless Power Transfer **Supplier: Fair-Rite**

As the automotive industry shifts from mechanical to electrical systems, including future addition of high-power Wireless Power Transfer (WPT) systems has created significantly more electromagnetic interference (EMI). This presentation introduces innovative ferrite solutions engineered to meet these challenges, focusing on enhanced EMI suppression and WPT efficiency. We will explore the critical role of ferrite in systems governed by the SAE standard J2954 for WPT and the growing EMI being generated from increased usage of power used in vehicles.

Passive Component Best Practices, Design Considerations, and Selection Guidance **Supplier: NIC**

Passive components, such as capacitors, resistors, and inductors, play a critical role in electronic design, accounting for up to 70% of the components on a PCB. Their proper selection ensures reliability, efficiency, and performance, particularly in applications like connectivity, energy, robotics, and industrial automation. High-performance capacitors (e.g., MLCCs, aluminum electrolytic capacitors, supercapacitors), precision resistors, and molded power inductors are essential for optimizing material utilization, reducing design size, and enhancing overall functionality.

Multimodal Intelligent Edge AI Interface Out-of-Box Solution **Supplier: Nuvoton**

This session shows how multimodal edge AI—gesture, vision, and voice—can run entirely on a microcontroller to enable touchless, intelligent human-machine interfaces. By performing inference locally, the solution delivers deterministic low latency, protects user privacy, and eliminates cloud dependency while maintaining cost efficiency.

Using a real-world implementation on the Nuvoton M55M1 MCU, the presentation demonstrates how multiple AI models are orchestrated through a streamlined development workflow and high-productivity tools. Attendees will gain insight into the system architecture, model workflow, and integration techniques needed to deploy responsive, AI-driven interfaces on resource-constrained devices, enabling autonomous, natural interaction across smart home, industrial, building, and medical applications.

A Practical Deep Dive into Real-World Image Sensor Design **Supplier: onsemi**

This session provides a practical guide to navigating imaging projects from start to finish, with support from onsemi and Arrow. Attendees will learn how to select the right image sensor based on key application requirements — including resolution, shutter type, dynamic range, lighting conditions, and interface constraints — while understanding the tradeoffs involved. The session will review onsemi's key image sensor families and their differentiating features, address common power supply and system integration challenges, and explore the broader imaging ecosystem available to reduce design risk, accelerate development, and shorten time-to-market.

Powering the Future with Toshiba **Supplier: Toshiba**

Discover Toshiba's latest advancements in Analog and Discrete semiconductors, featuring Silicon (Si), Silicon Carbide (SiC), and Gallium Nitride (GaN) technologies. These high-performance devices enable efficient power conversion and precise control in demanding industrial environments—from robotics and factory automation to motor drives and industrial inverters. Learn how Toshiba's innovations are shaping the future of reliable, scalable, and energy-efficient industrial solutions.

Wi-Fi HaLow: Long-Range IoT with km-Scale, mW-Power, Massive-Node Connectivity **Supplier: Vantron**

Wi-Fi HaLow (IEEE 802.11ah) carries Wi-Fi down into sub-1 GHz, delivering kilometer-range links that cut through walls and run for years on coin-cell power—yet still speak native IP, WPA3 security, and standard sockets. The presentation unpacks the technical levers that set HaLow apart: topologies scaling to thousands of nodes, duty-cycled power states that adapt to traffic, and modulation tuned for robust long-haul performance. Moving beyond the spec, we share best practices from field deployments—coexistence tricks, throughput-versus-distance trade-offs, and more. Use-case snapshots spanning industrial, connected-living, smart-factory, and city-scale infrastructure show how HaLow finally balances reach, security, and battery life. Attendees leave with a vendor-neutral, engineering-level playbook on where, when, and how to plug HaLow into next-gen designs to accelerate mass IoT roll-outs.

9:45 to 10:45 Classes

Power Customization for Specialized Applications and Architectures

Supplier: Advanced Energy

Applications in the medical and industrial sectors often feature unique architectures and demanding requirements that push design engineers to think creatively about their power solutions. Join Advanced Energy for a discussion on how power supplies can be adapted and customized to meet these specialized needs. This session will explore the types of applications and system architectures that drive customization, outline how power vendors collaborate with customers to fulfill these requirements, and highlight real-world case studies where tailored power solutions have enabled successful product designs.

Integrating and Qualifying Embedded Antennas: A System-Level Engineering Process

Supplier: Antenova

The seminar on integrating embedded antennas into multi-band wireless platforms focuses on critical aspects of antenna design and system integration. It emphasizes the importance of early technical engagement to align antenna design with system requirements and constraints. Key topics include managing design constraints such as size, material properties, and frequency bands, alongside best practices for PCB and mechanical integration to ensure optimal performance. The seminar also addresses common pitfalls like impedance mismatches and interference, offering mitigation strategies such as proper grounding and isolation techniques. Validation methods, including Over-The-Air (OTA) testing and system-level metrics like Total Radiated Power (TRP), Total Isotropic Sensitivity (TIS), and Effective Isotropic Radiated Power (EIRP), are discussed to ensure compliance and performance optimization.

Unlocking the Power of Riedon Resistors. Technologies, Capabilities, and Real-World Applications

Supplier: Bourns

This session offers a focused technical overview of Riedon's high-performance resistor technologies—now part of the Bourns portfolio—and how they solve demanding design challenges across industrial, medical, automotive, and instrumentation applications. Attendees will learn the advantages of Riedon's precision wire-wound, thick-film, thin-film, and power resistors, including their accuracy, thermal stability, pulse handling, and long-term reliability. The presentation highlights common engineering use cases such as improving sensing accuracy, managing high current and pulse loads, and ensuring stable performance in harsh or compact environments. Participants will gain practical selection guidance to help choose the right Riedon resistor technology to boost system robustness and accelerate reliable product design.

Advancing Smart Infrastructure with an XBee Wi-SUN Solution: A Unified Edge-to-Cloud Platform

Supplier: Digi

As cities, utilities, and industrial operators scale connected infrastructure, reliable and secure networking is essential. This session explores how the Digi XBee Wi-SUN Solution, built on the Silicon Labs EFR32 platform, enables standards-based sub-GHz mesh networking optimized for large-scale smart infrastructure deployments.

Attendees will learn how Digi XBee modules and the XBee Hive Border Router deliver long-range, resilient connectivity with edge intelligence and seamless enterprise and cloud integration. The session also highlights how Digi Remote Manager simplifies provisioning, security, monitoring, and over-the-air updates—reducing complexity, improving visibility, and accelerating smart city, utility, and industrial IoT rollouts.

How to Select the Right Ethernet Connectivity Technology for Industrial Applications

Supplier: Harting

Industrial Ethernet is the foundation of modern industrial communication, enabling reliable integration of devices across diverse environments. This session provides a concise overview of Ethernet connectivity options, comparing copper and fiber solutions, including traditional 4- and 8-wire copper versus emerging Single Pair Ethernet (SPE). Attendees will gain practical guidance on selecting the right cabling and connectivity technologies based on application requirements, environmental constraints, and scalability needs, with insights into where SPE and fiber-optic Ethernet deliver the most value in today's industrial systems.

Utilizing FPGA's as Companion Chips in Physical AI (sensor, camera, security) applications

Supplier: Lattice

As edge and physical AI applications push MCUs and MPUs to their performance and power limits, this session explores how ultra-low-power FPGAs can act as companion devices to offload functions such as sensor aggregation, camera interfacing, I/O expansion, and hardware-based security. Using solutions from Lattice Semiconductor, designers can add flexible acceleration and deterministic low-latency processing without redesigning their core processor architecture, extending system headroom and enabling new capabilities. The presentation highlights how compact FPGAs can scale edge AI platforms and extend the lifecycle of existing processors from vendors including Qualcomm, NXP, STMicroelectronics, and NVIDIA—with minimal redesign effort.

Configurable Logic - Harnessing the Power of Programmable Logic in PIC® Microcontrollers

Supplier: Microchip

Enhance your applications by integrating complex logic designs using the Configurable Logic Block (CLB) on PIC® microcontrollers. You will explore the CLB architecture, including Look-Up Tables (LUTs), interconnections, sequential logic, storage elements, and clocking. Additionally, you will learn to use the CLB Logic configuration tool, including schematic capture, library elements, and circuit hierarchy.

11:00 to 12:00 Classes

Solve common design problems with programmable, mixed-signal uASiCs

Supplier: Diodes Inc

Learn how to save board space, power, and costs with Diodes' configurable uASiCs solutions for industrial and consumer applications. In this session we will address common design needs like power sequencing, watchdog supervisors, waveform generators, voltage monitors, OVP/UVP and OCP functions, and the implementation of simple and complex logic functions in a programmable, mixed-signal ASiC. We will show design tool examples and demonstrate how simple it is to implement a solution using uASiCs from Diodes Inc.

Discover How Advanced Pressure & Flow Sensing Can Transform Your Applications

Supplier: Honeywell

Join us for an engaging session exploring how next-generation pressure and liquid flow sensors are powering innovation across medical, industrial, and aerospace markets. These sensors play a vital role in the accuracy, safety, and reliability of critical systems—from life-saving medical devices to high-performance industrial and aerospace equipment.

You'll see how our sensing technologies enable smarter, more dependable operation in applications such as dialysis equipment, blood analysis systems, centrifuges, oxygen and nitrogen gas distribution, HVAC units, data storage, process controls, industrial machinery, pumps, and robotics.

Passive Component Solutions for Factories of the Future

Supplier: KAVX

Industrial controllers rely on precise timing and robust power quality, and advancements in timing devices and passive components are reshaping performance on the factory floor. This session highlights how next-generation crystals, oscillators, and TCXOs deliver the accuracy, stability, and reliability needed across processing boards, distributed systems, robotics, and HMI platforms. We will also examine recent improvements in capacitors and multilayer varistors that support size-, weight-, and power-optimized designs, along with guidance on selecting the right technologies for varying industrial power demands. Attendees will gain practical insight into how modern timing and passive solutions improve system resilience, maintain performance under harsh conditions, and enable more reliable, future-ready industrial controllers.

Timekeeping via RTC Modules – Technologies and Applications for Modern Devices

Supplier: Micro Crystal

This seminar explores the latest advancements in ultra-low power timekeeping technologies and their essential role in today's energy-constrained electronic systems. Attendees will gain a comprehensive understanding of how Micro Crystal's RTC and oscillator solutions deliver new levels of efficiency, precision, and scalability — enabling more sustainable designs across a broad range of modern applications.

Architecting EdgeAI Systems with the i.MX 95: A Deep Dive for Embedded Engineers

Supplier: NXP

This session provides a technical deep dive into NXP's i.MX 95 applications processor family, covering system architecture, subsystem behavior, and key design considerations for AI-enabled embedded platforms. Engineers will learn how to evaluate the i.MX 95's capabilities, leverage its heterogeneous processing elements and accelerators, and integrate AI/ML features into production-grade designs for industrial automation, automotive edge, medical, commercial IoT, and connected embedded applications. The presentation also positions i.MX 95 within NXP's broader secure and scalable EdgeAI portfolio—from ultra-low-power MCUs to high-performance processors and discrete NPUs delivering up to 40 eTOPS—providing a unified path for deploying intelligent systems across diverse performance and power requirements.

Enable Accurate Distance Estimation Using Channel Sounding

Supplier: Silicon Labs

Explore the latest advancements in Bluetooth 6.0, with a special focus on Channel Sounding. Channel Sounding is a new standardized, interoperable, and secure method for distance measurement that is set to revolutionize a variety of applications. This session will also feature a comparative analysis of Channel Sounding with other ranging methods.

Advancing Power with Next-Gen SiC for Medical, Industrial, and High-Reliability Systems

Supplier: Wolfspeed

This session introduces Wolfspeed as the industry's leading vertically integrated SiC power semiconductor provider, showcasing their newest discrete devices and power modules. Attendees will discover how next-generation SiC enables higher efficiency, faster switching, and greater power density across demanding applications — with a focus on medical pulse-power, high-reliability power systems, motor drives, and industrial power supplies. The session concludes with a practical design walkthrough using Wolfspeed's SpeedFit™ online simulator and SpeedVal™ evaluation platform, demonstrating how SiC technology can accelerate development and reduce design risk.

1:00 to 2:00 Classes

Solving actual problems with EdgeAI

Supplier: Alif Semiconductor

This training session highlights Alif Semiconductor's Ensemble family of microcontrollers, delivering up to two orders of magnitude improvements in Edge-AI performance and power efficiency compared to traditional MCUs. By combining advanced compute architectures with ultra-low-power design, the Ensemble family enables always-on intelligent sensing and inference at the edge, unlocking new AI-driven applications within tight power and cost constraints. The session also introduces the Balletto family with BLE 5.3+ connectivity, showcasing real-world use cases that pair high-efficiency Edge-AI with secure, low-power wireless communication for smart devices, wearables, industrial sensing, and connected embedded systems.

Low Precision Math Support in FPGAs – Useful for AI and More!

Supplier: altera

This session explores how Altera FPGAs natively support low-precision math operations through the integration of Tensor math blocks and complex multipliers directly within DSP blocks. Attendees will learn how Tensor-optimized FIR and FFT implementations can significantly reduce device resource utilization, making them ideal for AI inferencing and high-performance DSP workloads. The session will also cover how MATLAB can be leveraged as an end-to-end tool — from designing efficient RTL to validating designs directly in hardware — and will conclude with a live demonstration.

Got HMI? We can Light the Way!

Supplier: ams OSRAM

This session explores how to modernize Human-Machine Interfaces using ams OSRAM's latest high-resolution multi-zone dToF sensors, optical force-sensing technology, impedance-based sensing, and non-contact position-sensing solutions. Attendees will learn how these advanced sensing modalities enable sealed, reliable, and intuitive touch-free interfaces suitable for harsh or hygienic environments. Complemented by a broad portfolio of light-emitting devices for visual feedback, these technologies allow engineers to build next-generation HMIs that are responsive, robust, and visually engaging. Join us to discover how ams OSRAM can help you bring your user interface designs up to modern standards. Let us help you – Light The Way.

Making Sense of Sensing: A Guide to Magnetic and Thermal Sensor Technologies

Supplier: Littelfuse

Join us for a focused technical session on magnetic and thermal sensing technologies used across modern electronic systems. This seminar provides engineers with a clear, practical understanding of Reed, Hall Effect, TMR, NTC thermistors, and RTDs—including how they work, when to use them, and how to optimize system performance, reliability, and cost. Real-world application examples will highlight design considerations such as sensitivity, power consumption, switching speed, environmental robustness, and integration trade-offs. Engineers looking to make smarter sensor selections and streamline their next design will gain immediate, actionable insights from this session.

Introducing Next-Gen Embedded Connectivity Platform: The Future of Embedded Cellular

Supplier: Multi-Tech

This seminar introduces a next-generation, 5G-ready embedded cellular connectivity platform designed for long-term IoT deployments across smart infrastructure, industrial IoT, and transportation. The platform simplifies cellular integration with automatic network management, a single global radio, and a Wi-Fi-like software workflow, while improving scalability, reliability, and time to market. The session also covers a practical, host-controlled approach to firmware over-the-air updates, giving developers greater control, reduced risk, and a faster, more cost-effective path to adding secure cellular connectivity with confidence.

How to Overcome Connectivity Challenges for Robotic Integration

Supplier: Phoenix Contact

Robotics relies on robust connectivity for real-time control, data exchange, and system integration. This encompasses internal electrical and electronic connections as well as communication between robots and external devices. Performance is often constrained by latency, bandwidth limitations, and network instability—particularly in autonomous vehicles, industrial automation, and multi-robot environments. Advances in connector design and emerging technologies offer improvements, yet achieving reliable, low-latency connectivity in dynamic conditions remains a critical challenge for fully autonomous and collaborative robotics.

High-Voltage DC-DC Power Design: Efficient Buck and Iso-Buck Architectures up to 100 V

Supplier: STMicro

Rugged 12 V, 24 V, 48 V, and up to 100 V power buses demand DC-DC solutions that are efficient, compact, and reliable. This session explores how modern buck and iso-buck topologies using STMicroelectronics' DCPx DC-DC Buck Converters simplify high-voltage power design through advanced control methods such as V2COT, light-load optimization for low power or low noise, and high-frequency operation that reduces magnetics size while maintaining stability and EMC performance. Attendees will also learn how iso-buck architectures enable compact, transformer-isolated supplies without optocouplers, and how tools like ST eDesignSuite and ST eDSim accelerate evaluation of efficiency, loop response, and robustness for demanding industrial and automotive applications.

2:15 to 3:15 Classes

North American & Global Market Update

Supplier: Arrow

North American and global economic view of the electronics market, including production and lead times, impacts of geopolitical events, and other factors affecting the supply chain.

Supercharge Your Power Designs: Mastering High-Reliability Sizing, Selection, and Tradeoffs

Supplier: Eaton

Engineers looking to boost power design reliability will receive practical guidance on sizing and selecting Eaton protection devices, magnetic inductors, and supercapacitors. This session breaks down key tradeoffs — including thermal limits, transients, EMI/EMC, lifetime, and cost — equipping attendees with clear frameworks, real-world examples, and actionable tips to make smarter component choices, avoid common design pitfalls, and build more robust, efficient power systems.

The Future of Connectivity: ACW741x Wi-Fi 7 and Bluetooth 6.0 Powering Next-Gen IoT Devices

Supplier: Infineon

The next phase of smart home and enterprise technology requires more than just speed—it requires, reliability, and low power consumption simultaneously. This presentation examines the convergence of Wi-Fi 7 and Bluetooth 6.0, illustrating how these upcoming technologies will redefine user experiences in 2026 and beyond. We will discuss how Wi-Fi 7, with its capacity to handle dense environments and massive data rates, pairs with the low-power efficiency and precise positioning capabilities of Bluetooth 6.0 to create seamless "smart" experiences. The session will cover key benefits, including near-instantaneous streaming, enhanced IoT-20 (Internet of Things in 20MHz channels) device communication, and improved battery life for peripheral devices. Attendees will gain actionable insights into how to leverage this technology to future-proof their product lines and networks.

Power Management Solutions for Energy Efficient Power System Designs

Supplier: Panasonic

This Panasonic seminar highlights how the right electronic components can make—or break—designs in demanding, real-world environments. As applications push into harsher and more compact operating conditions, engineers must contend with elevated temperatures up to +175 °C, protection against corrosive sulfurization, vibration levels of 30 G's or more, and high-voltage switching in constrained spaces. This session introduces component technologies purpose-built to withstand these challenges, helping designers improve reliability, extend product lifetimes, and avoid costly field failures. Because when components fail, customers notice—and Panasonic is here to make sure your designs don't.

Elevating UX Design with Audio, Haptics, & Sensors

Supplier: PUI

This session explores how audio, haptics, and sensing technologies can elevate user experience across medical, industrial, security, consumer, and point-of-sale systems by treating sound and tactile feedback as core design elements rather than afterthoughts. Attendees will learn how to evaluate and integrate speakers, piezo devices, haptics, and microphones based on environmental, space, and power constraints, and how intentional audio design improves usability, safety, and intuitive interaction—especially in rugged or noisy environments. The seminar also introduces essential acoustic principles, such as enclosure effects, resonance, and frequency response, and uses real-world examples to show how thoughtful audio and haptic integration can enhance system performance, user engagement, and product differentiation.

Moving AI to the Edge- and Making it Easier

Supplier: Qualcomm

Learn how Qualcomm is empowering the IoT ecosystem with advanced on-device AI capabilities using the Qualcomm AI Engine. Designed to run machine learning models directly on IoT devices, the platform enables faster local inference while enhancing privacy and security by minimizing reliance on the cloud. By reducing network bandwidth requirements and latency, the Qualcomm AI Engine supports more responsive, efficient, and scalable edge intelligence, helping developers deploy smarter, more autonomous connected devices across a wide range of IoT applications.

A New Era of High-Performance Timing Solutions

Supplier: Skyworks

This session examines the latest trends in clock tree design across key industry standards including PCIe and Ethernet, and how Skyworks timing solutions help engineers achieve optimal system performance. Attendees will learn how to leverage Skyworks tools to meet the precise timing requirements of modern architectures, and explore reference design platforms spanning FPGAs, switch SoCs, embedded processors, and data center applications that utilize Skyworks timing products.